Python data science (continued)

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Tutorial #3: String operations

- pandas-cookbook: String operations (Chapter 6)
  - This tutorial is about extracting information from text in `pandas`.
  - *Hint: Some of the most interesting information in tomorrow's dataset might be in string variables.*
- Core concepts:
  - Detecting keywords in strings:

```
1 weather_description.str.contains('Snow')
```

Tutorial #4: Data cleaning

- pandas-cookbook: Data cleaning (Chapter 7)
- Core concepts:
  - Detecting `nan` values stored as strings:

```
na_values = ['NO CLUE', 'N/A', '0']
requests = pd.read_csv(..., na_values=na_values, ...)
```

  - Altering `DataFrame` values in-place:

```
zero_zips = requests['Incident Zip'] == '00000'
requests.loc[zero_zips, 'Incident Zip'] = np.nan
```

Tutorial #5: sklearn - Linear Regression

- sklearn - Linear Regression Example
- Core concepts:
  - Fitting a model to data

```
regr.fit(diabetes_X_train, diabetes_y_train)
```

  - Making predictions with a model

```
regr.predict(diabetes_X_test)
```

  - Model evaluation

```
mean_squared_error(diabetes_y_test, diabetes_y_pred)
```

Challenge scaffold

- challenge_scaffold.py
  - Look at the Challenge Guidelines and the project scaffold.
  - I will walk through the challenge scaffold!
  - Use this scaffold as a starting point for the challenge tomorrow.

Extra

- Start working with the dataset we will use tomorrow.
- Work on a few of the items in the challenge:
  - *What variables are in the dataset?*
  - *What are the data types of the variables?*
  - *Is there any missing data?*